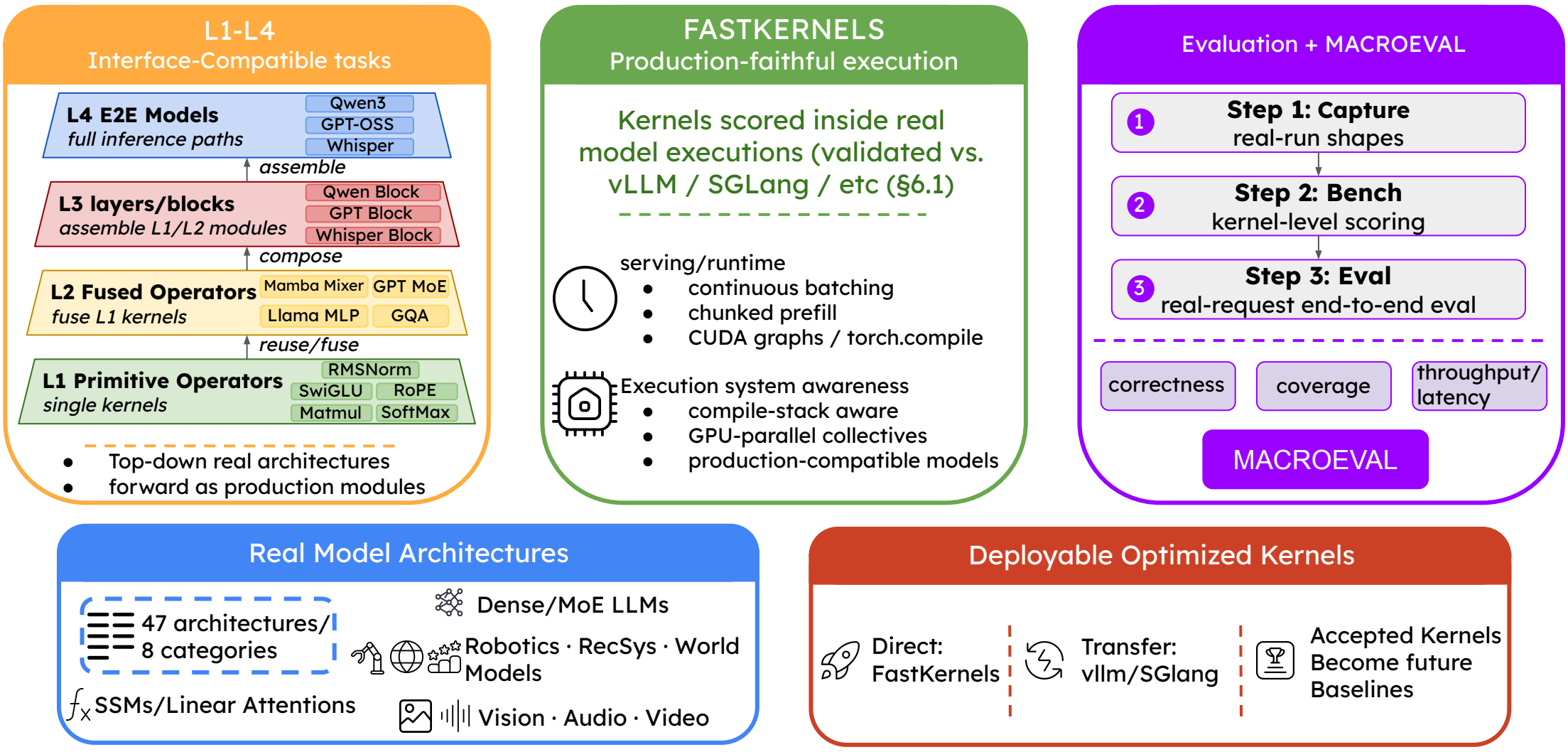

L1-L4
Interface-Compatible tasks
L4 E2E Models
full inference paths
Qwen3
GPT-OSS
Whisper
assemble
L3 layers/blocks
assemble L1/L2 modules
Qwen Block
GPT Block
Whisper Block
compose
L2 Fused Operators
fuse L1 kernels
Mamba Mixer
GPT MoE
Llama MLP
GQA
reuse/fuse
L1 Primitive Operators
single kernels
RMSNorm
SwiGLU
RoPE
Matmul
SoftMax
Top-down real architectures
forward as production modules
FASTKERNELS
Production-faithful execution
Kernels scored inside real model executions (validated vs. vLLM / SGLang / etc (§6.1)
serving/runtime
continuous batching
chunked prefill
CUDA graphs / torch.compile
Execution system awareness
compile-stack aware
GPU-parallel collectives
production-compatible models
Evaluation + MACROEVAL
1
Step 1: Capture
real-run shapes
2
Step 2: Bench
kernel-level scoring
3
Step 3: Eval
real-request end-to-end eval
correctness
coverage
throughput/ latency
MACROEVAL
Real Model Architectures
47 architectures/ 8 categories
Dense/MoE LLMs
Robotics · RecSys · World Models
SSMs/Linear Attentions
Vision · Audio · Video
Deployable Optimized Kernels
Direct: FastKernels
Transfer: vllm/SGlang
Accepted Kernels Become future Baselines